Applications of Data Science: LLMs

2. Understanding Large Language Models

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Applications of Data Science: LLMs

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

1st class - Understanding Language Models (11 March)

2nd class - Understanding Large Language Models (18 March)

3rd class - Training Large Language Models (25 March)

4th class - Attention mechanism (8 April)

5th class - Transformer architecture (15 April)

6th class - Presentations of the final projects (22 April)

- Assignment (before the last class on 22 April):
  - Submit your final project's notebooks/data/reports to the course GitHub repository

Overview

- About LM
  - Theory/Intuition
    - White-board
  - Ngram LM
    - Hands-on (Python)

- **Recap**
  - Quiz / Name check (SV)
  - Discuss quiz (QA)
  - Sampling (Demo)

- **About LLM**
  - ML/NN/Transformer

  - Inference
    - Input
      - Encoding (tokenization)
    - Output
      - Decoding (sampling)

- **Course Projects**

Quiz #1

- Canvas

Sampling (Demo)

n-gram Language Model

- approximate probability
- based on the preceding n-1 words
    - P(w4|w1, w2, w3)

- **bigram** model
    - only the previous word
        - ≈ P(w4|w3)

- 3-gram (**trigram**) model
    - two preceding words to predict the next word
        - ≈ P(w4|w2, w3)

Why LLM?

Why LLM?

- Long-range dependencies
  - 0-probabilities

- Relations between words
  - Synonyms

Large Language Model (LLM)

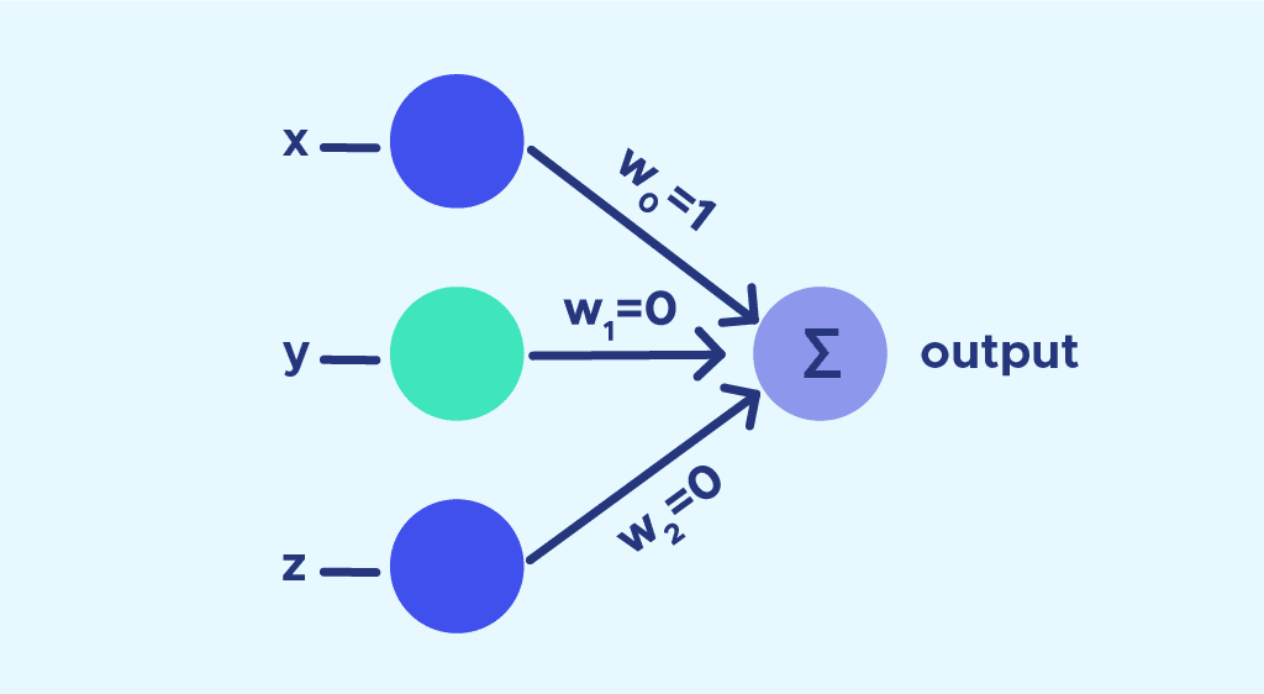
Large Language Model (LLM)

- neural network

Neural Network

- models a nonlinear **function**,
  - i.e., a relationship between input and output
- using learnable parameters (weights)

Neural Network (Perceptron)



https://datascientest.com/en/perceptron-definition-and-use-cases

Activation (Step) Function

$$z = \sum_{j=1}^{m} x_j w_j = w^T x$$

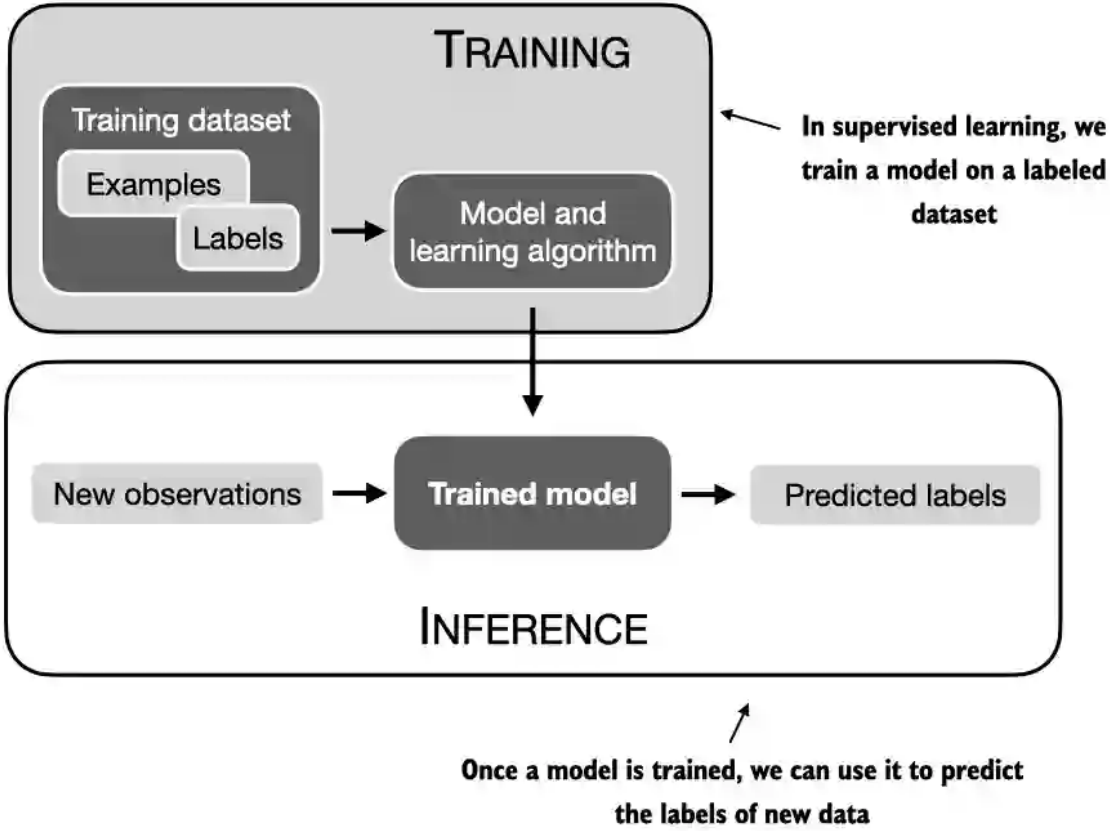
$$\mathrm{g(z)} = \begin{cases} 1, & \text{if } z \geq \theta \\ -1, & \text{otherwise} \end{cases}$$

https://medium.com/codex/single-layer-perceptron-and-activation-function-b6b74b4aae66

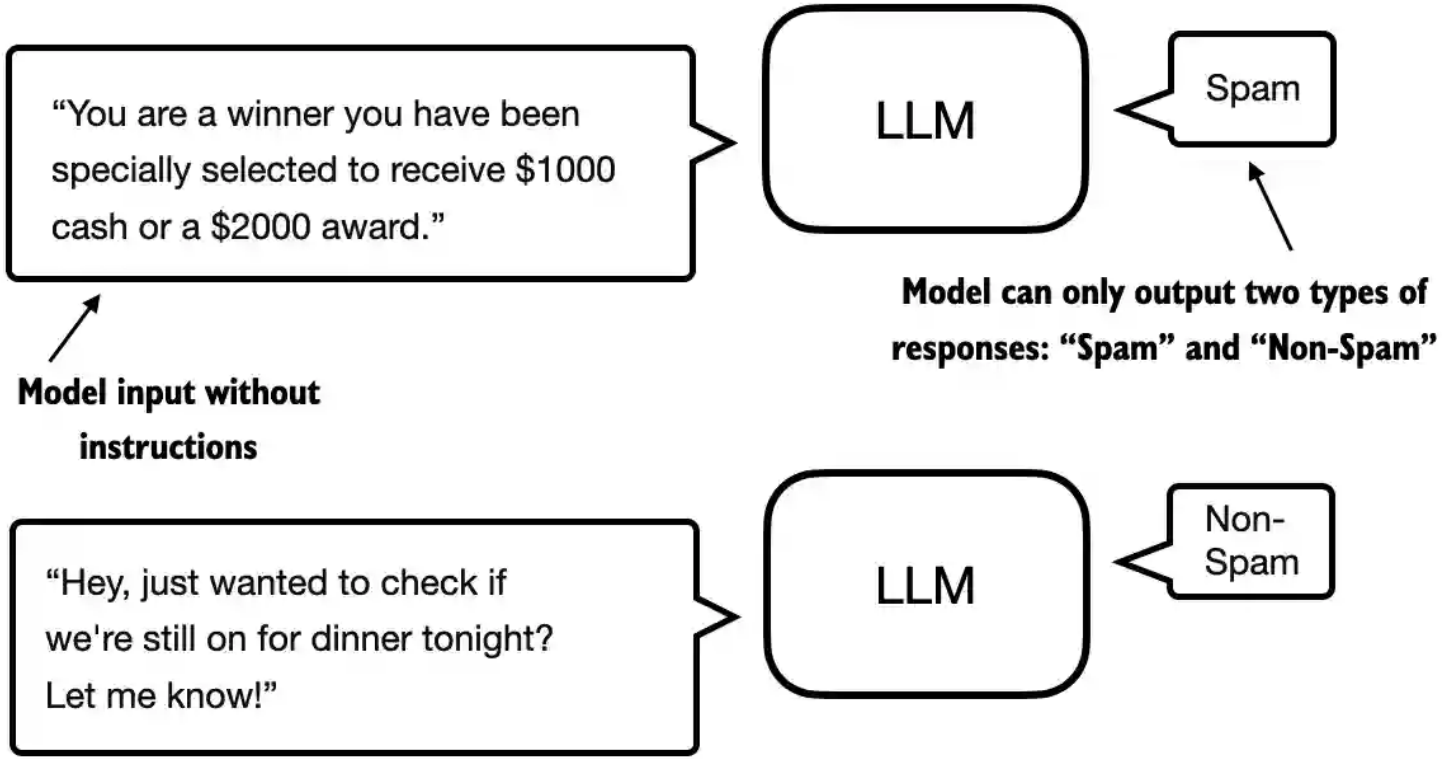
Large Language Model (LLM)

- neural network
  - **machine learning** model

Machine learning



Classification



Transformer

Transformer

- **deep neural network**
  - encoder-decoder architecture
  - "Attention Is All You Need" Google 2017

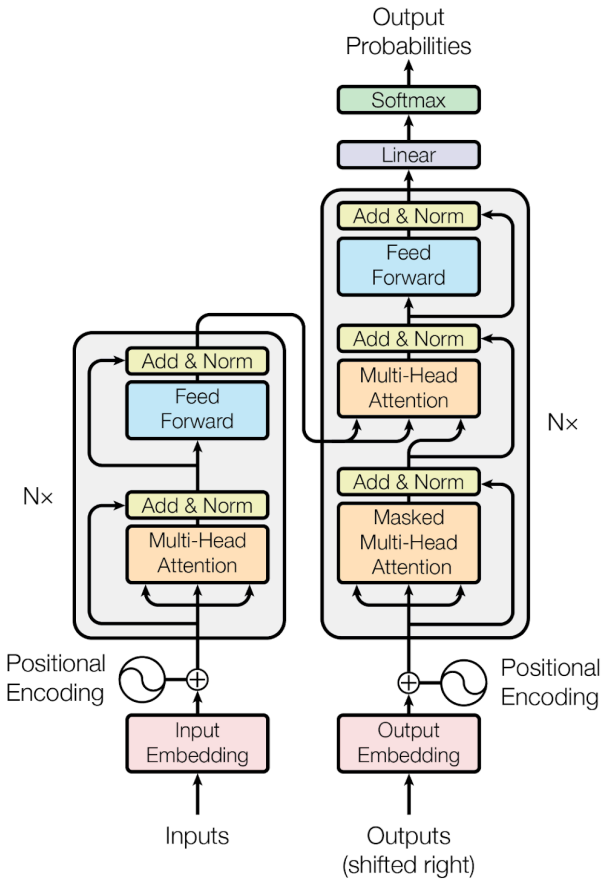
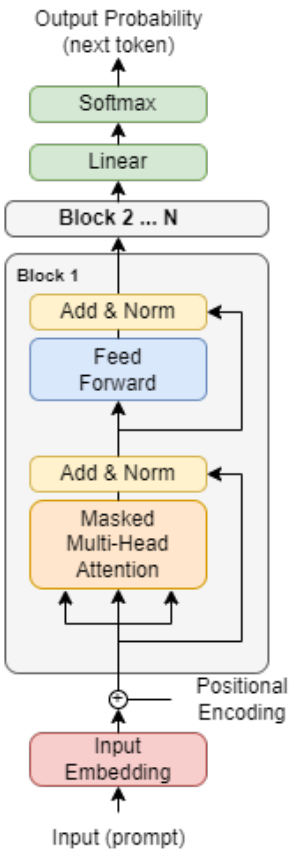


Figure 1: The Transformer - model architecture.

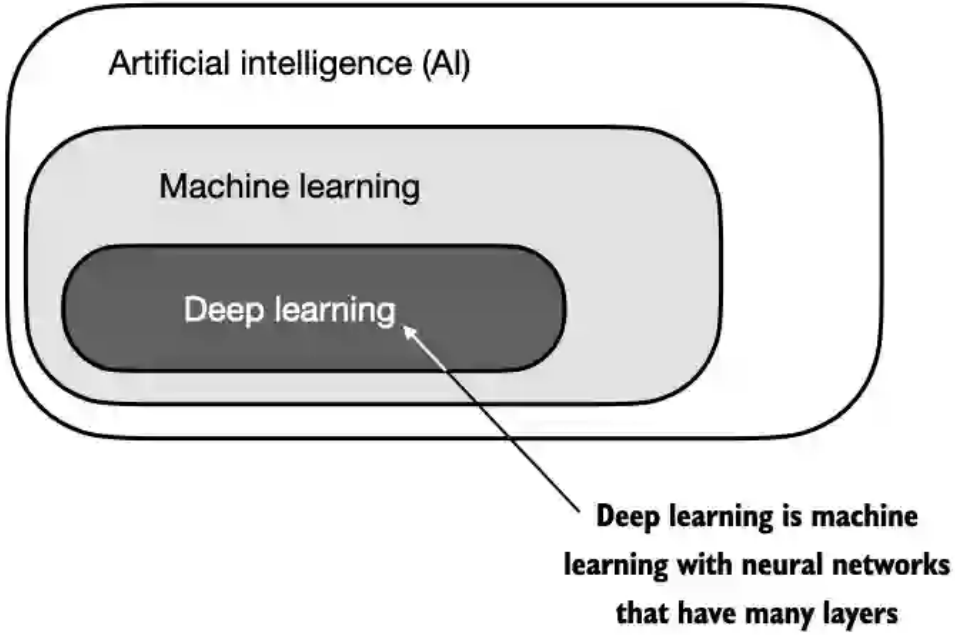
GPT

- deep neural network
  - **decoder-only** architecture

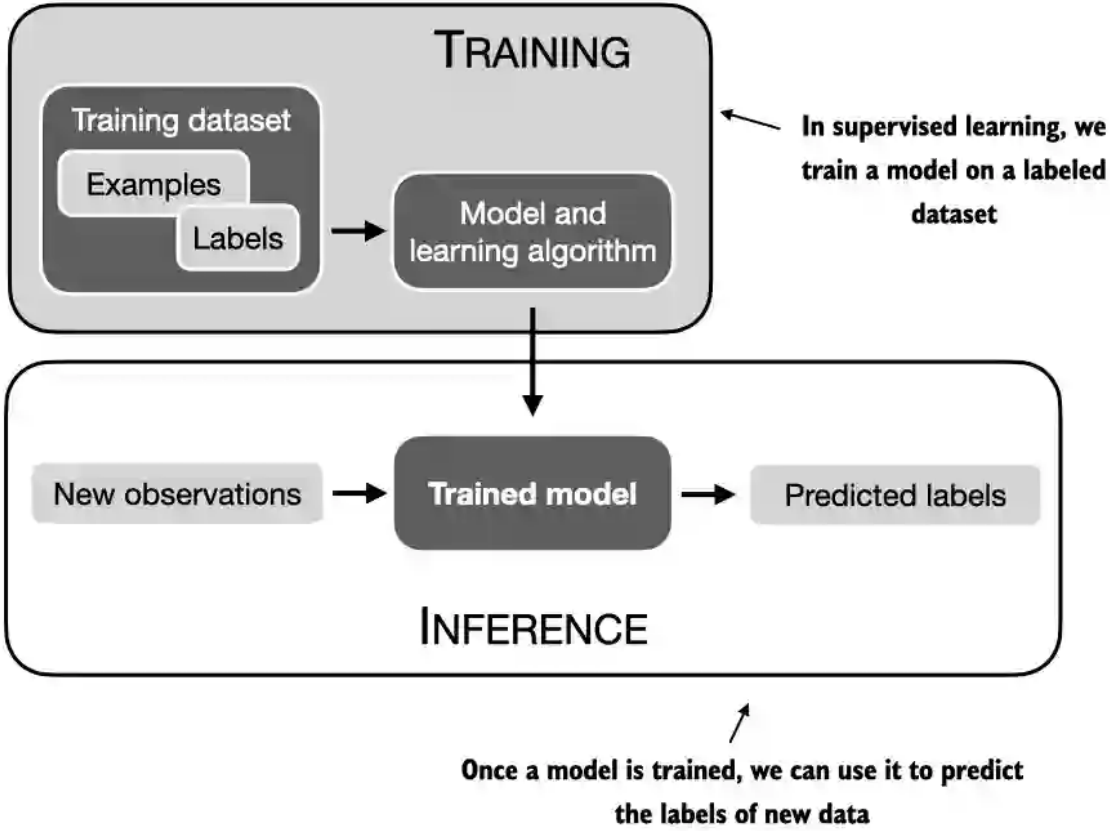


https://ai.stackexchange.com/questions/40179/how-does-the-decoder-only-transformer-architecture-work

Deep learning



Machine learning: Inference



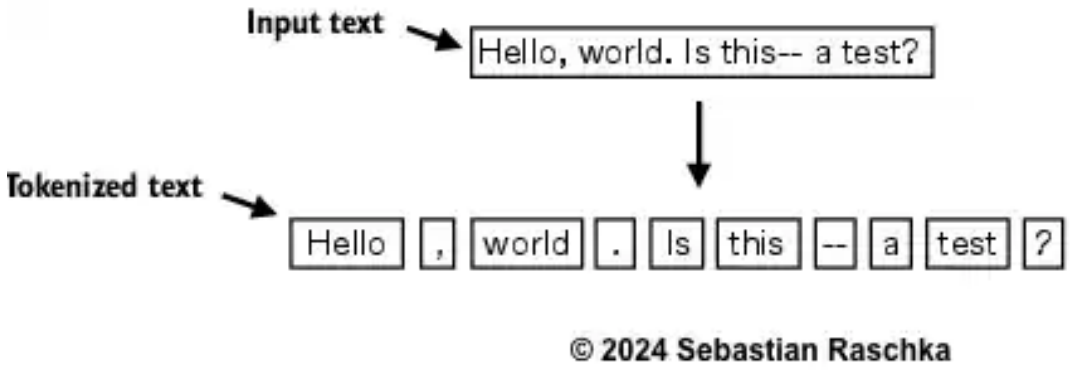
Transformer Input

Tokenization

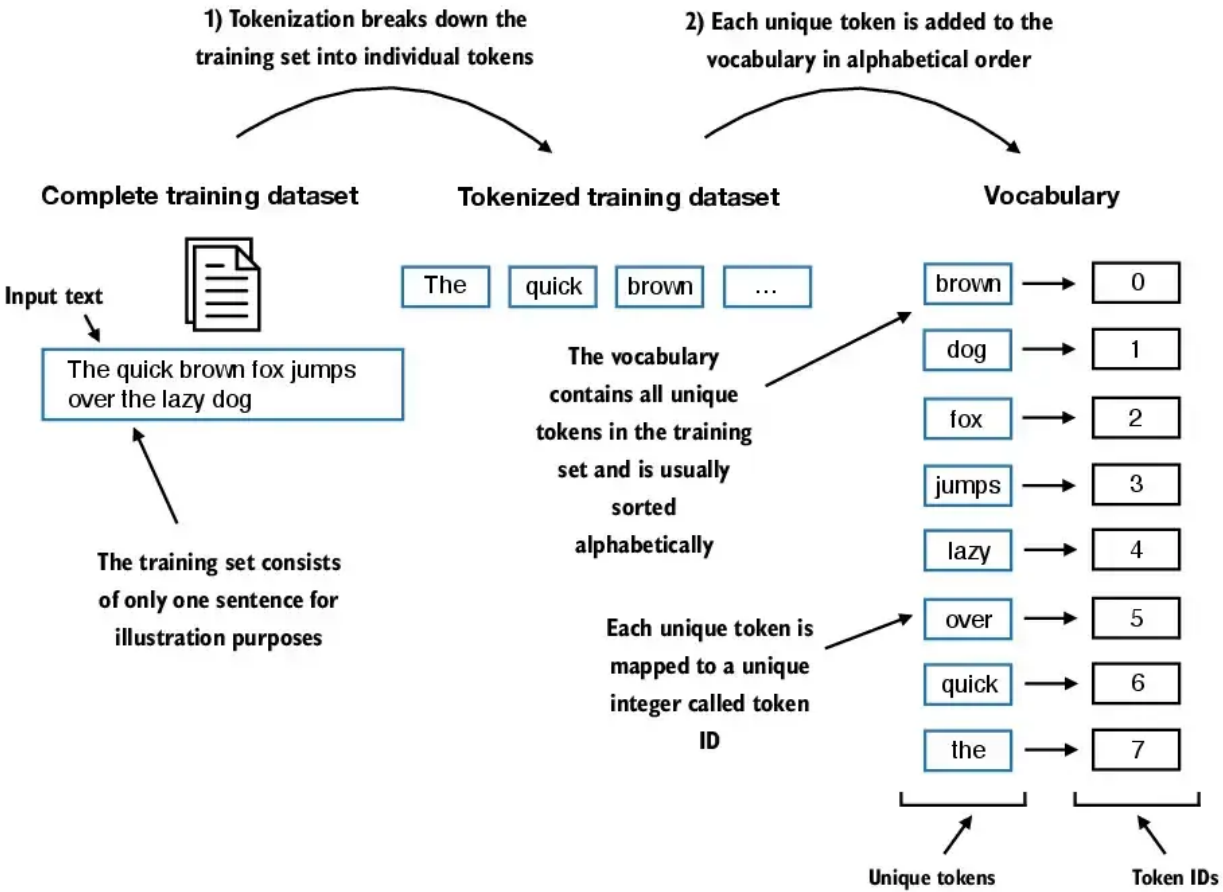
Tokenization

- break text into smaller parts
  - convert text into numbers

Tokenization



Token IDs



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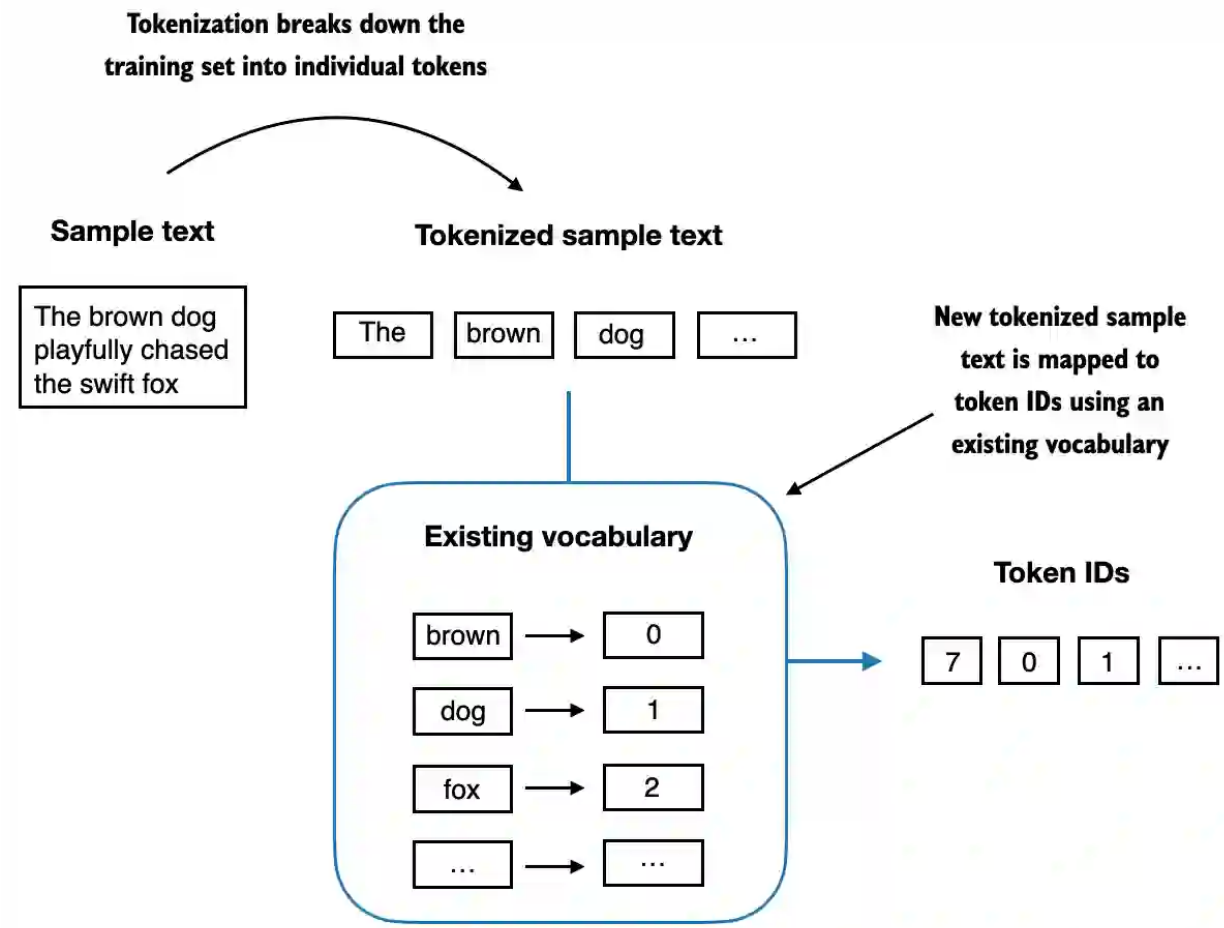
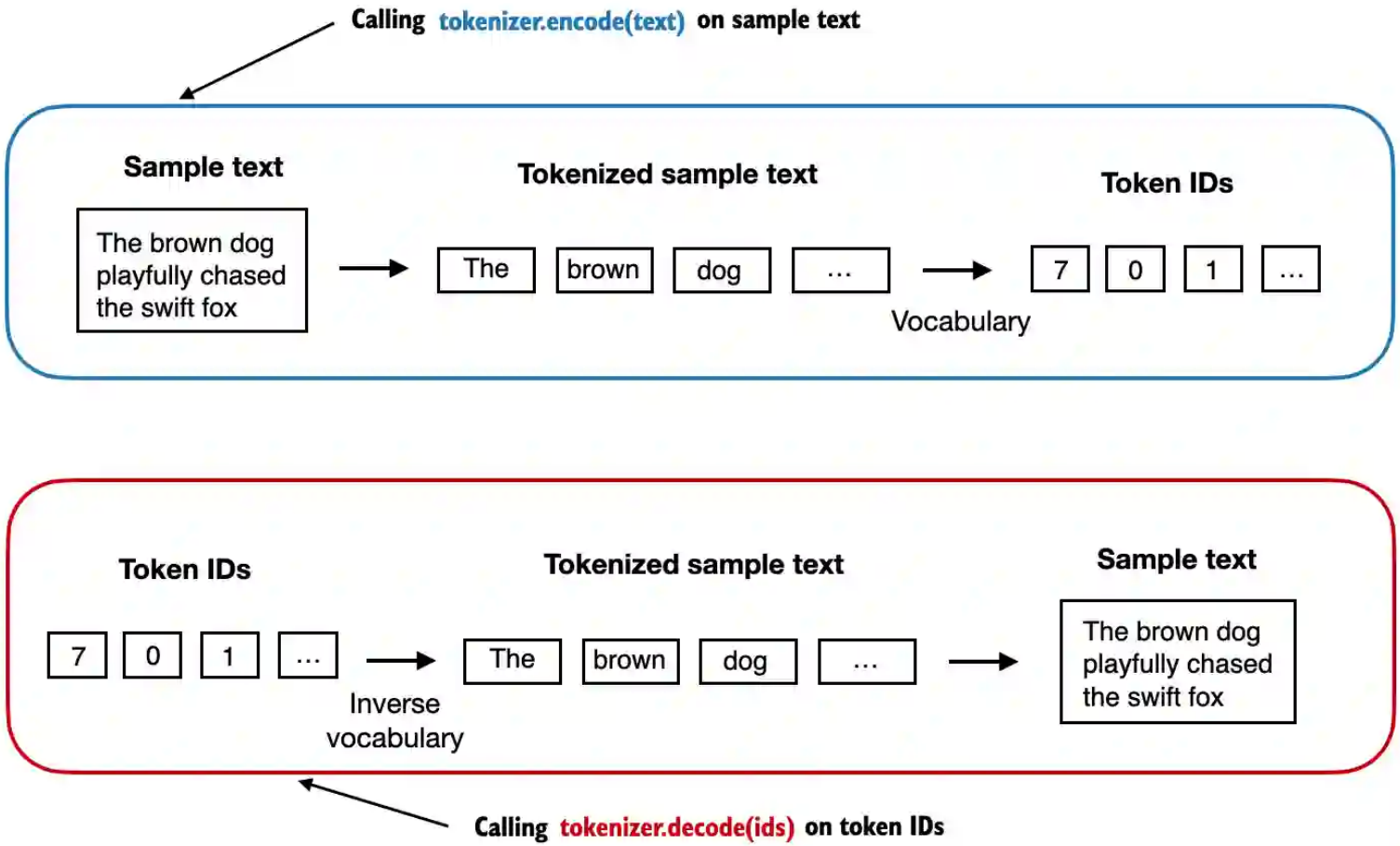
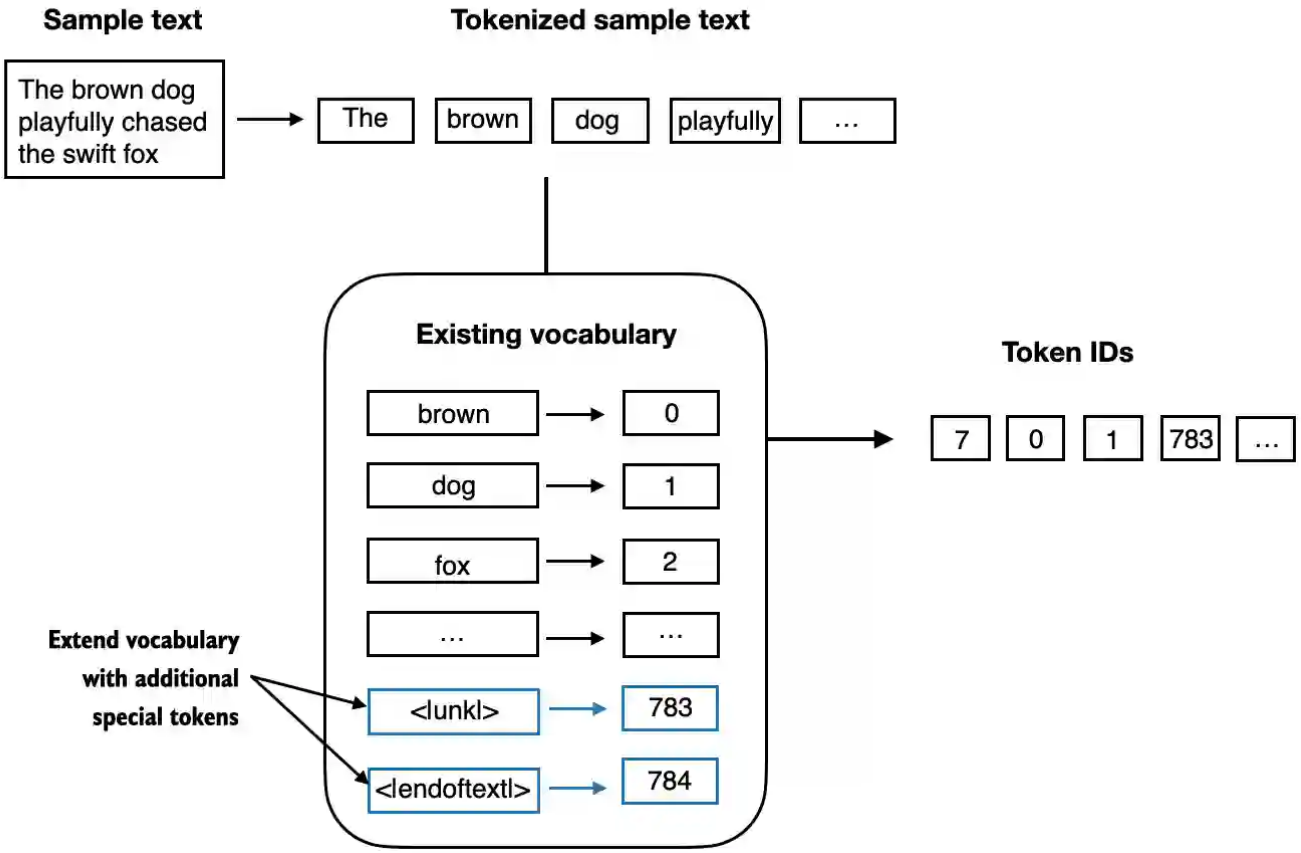
Encoding


Tokenization breaks down the training set into individual tokens
Sample text
The brown dog playfully chased the swift fox
Tokenized sample text
The
brown
dog
...
New tokenized sample text is mapped to token IDs using an existing vocabulary
Existing vocabulary
brown
0
dog
1
fox
2
...
...
Token IDs
7
0
1
...

Decoding


Calling tokenizer.encode(text) on sample text
Sample text
The brown dog playfully chased the swift fox
Tokenized sample text
The
brown
dog
…
Vocabulary
Token IDs
7
0
1
…
Token IDs
7
0
1
…
Inverse vocabulary
Tokenized sample text
The
brown
dog
…
Sample text
The brown dog playfully chased the swift fox
Calling tokenizer.decode(ids) on token IDs

Special tokens



Tokenization

- convert text into numerical data
  - break text into smaller parts

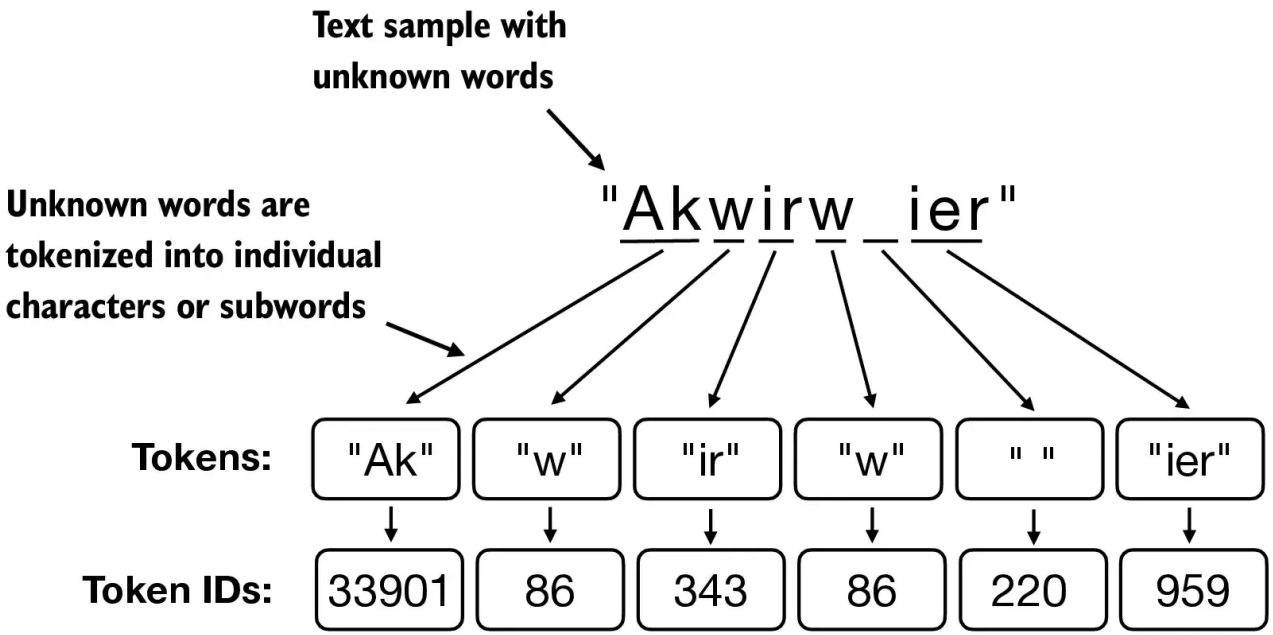
Tokenization

- convert text into numerical data
  - break text into smaller parts
    - words
    - characters

Tokenization

- convert text into numerical data
  - break text into smaller parts
    - words
      - large vocabulary
      - short code length
      - unknown (unseen) words
    - characters
      - small vocabulary
      - long code length
      - no out-of-vocabulary (unknown) tokens

Subwords



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Jupyter

Tensor

A scalar is just a single number

$$2$$

Scalar

0D tensor

An example of a 3D vector that consists of 3 entries

$$\begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$$

Vector

1D tensor

A matrix with 3 rows and 4 columns

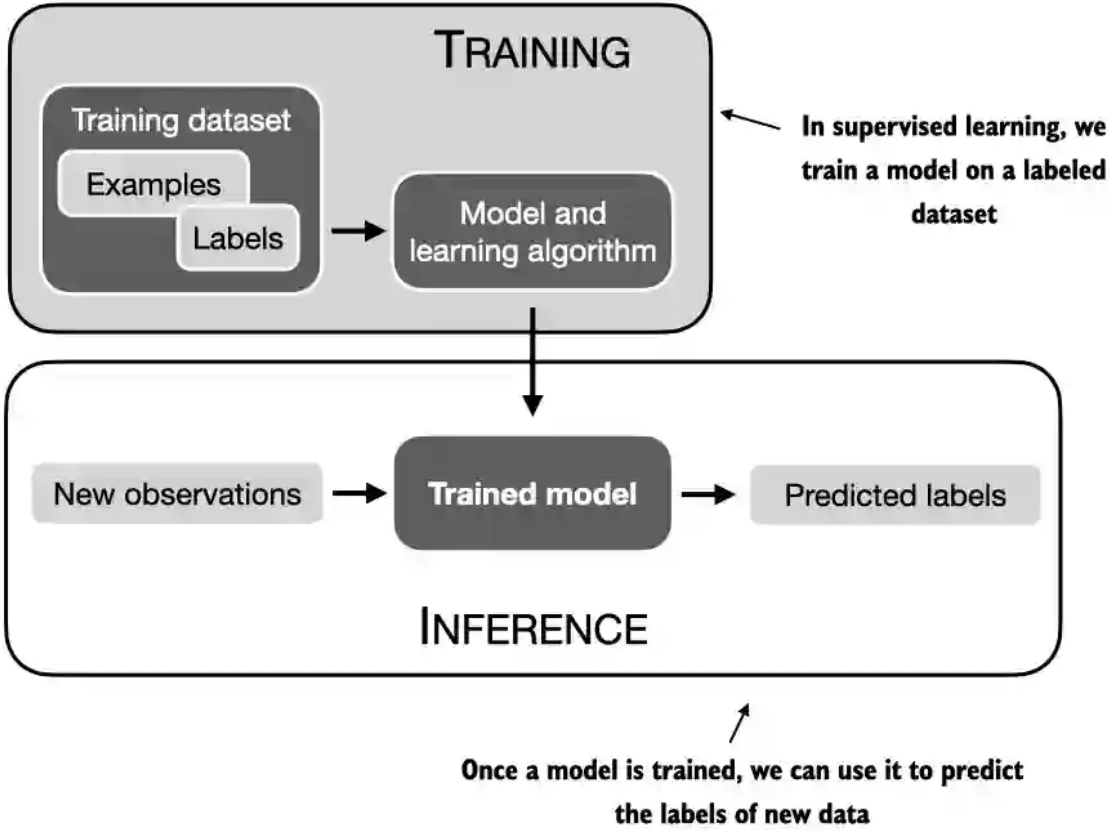
$$\begin{bmatrix} 3 & 5 & 1 & 2 \\ 1 & 7 & 2 & 3 \\ 3 & 3 & 4 & 9 \end{bmatrix}$$

Matrix

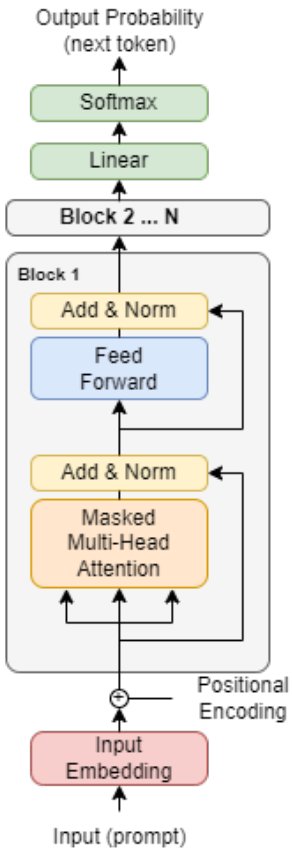
2D tensor

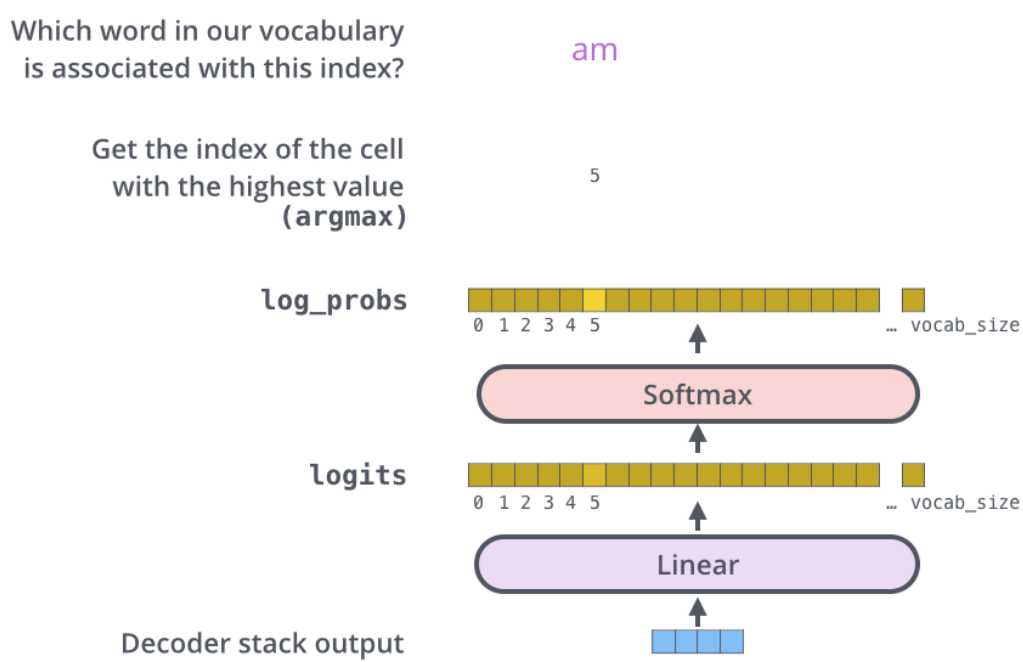
Jupyter

LLM Inference



Transformer Output





https://www.kaggle.com/code/colearninglounge/nlp-model-building-transformers-attention-more

Jupyter

- lecture_2_inference.ipynb

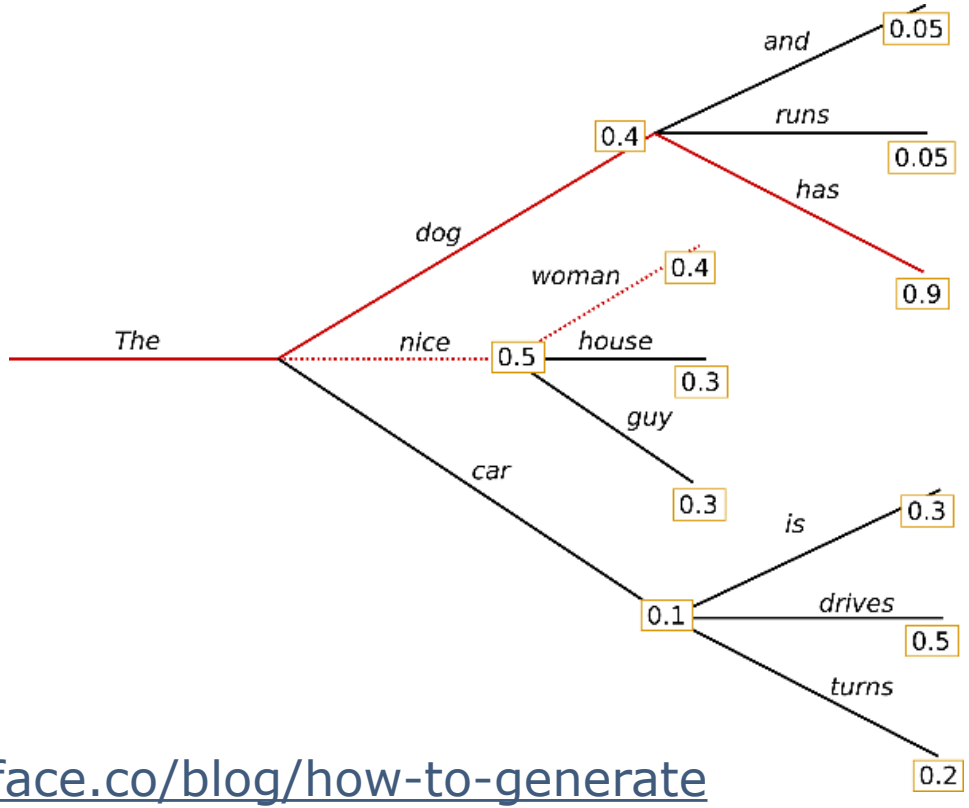
Decoding methods

Decoding methods

- Greedy decoding

Decoding methods

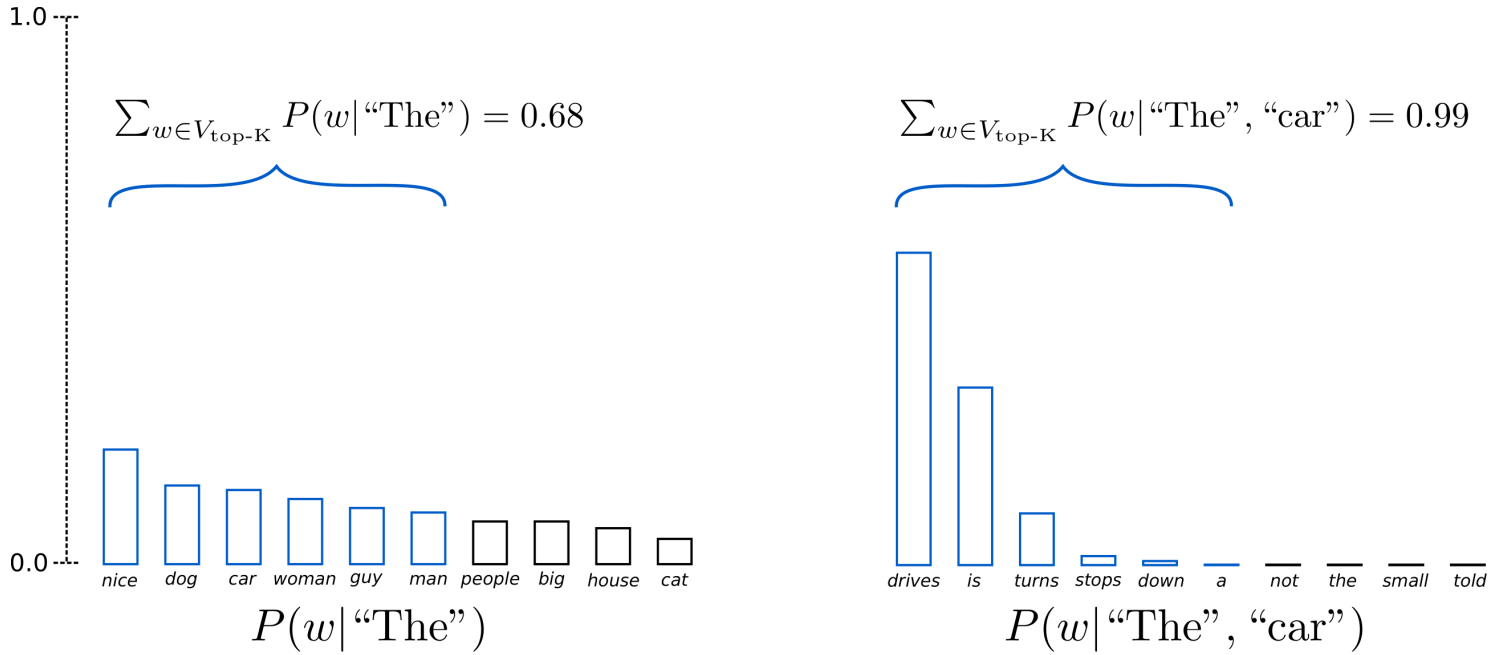
- Greedy decoding
- **Beam search (tree)**



https://huggingface.co/blog/how-to-generate

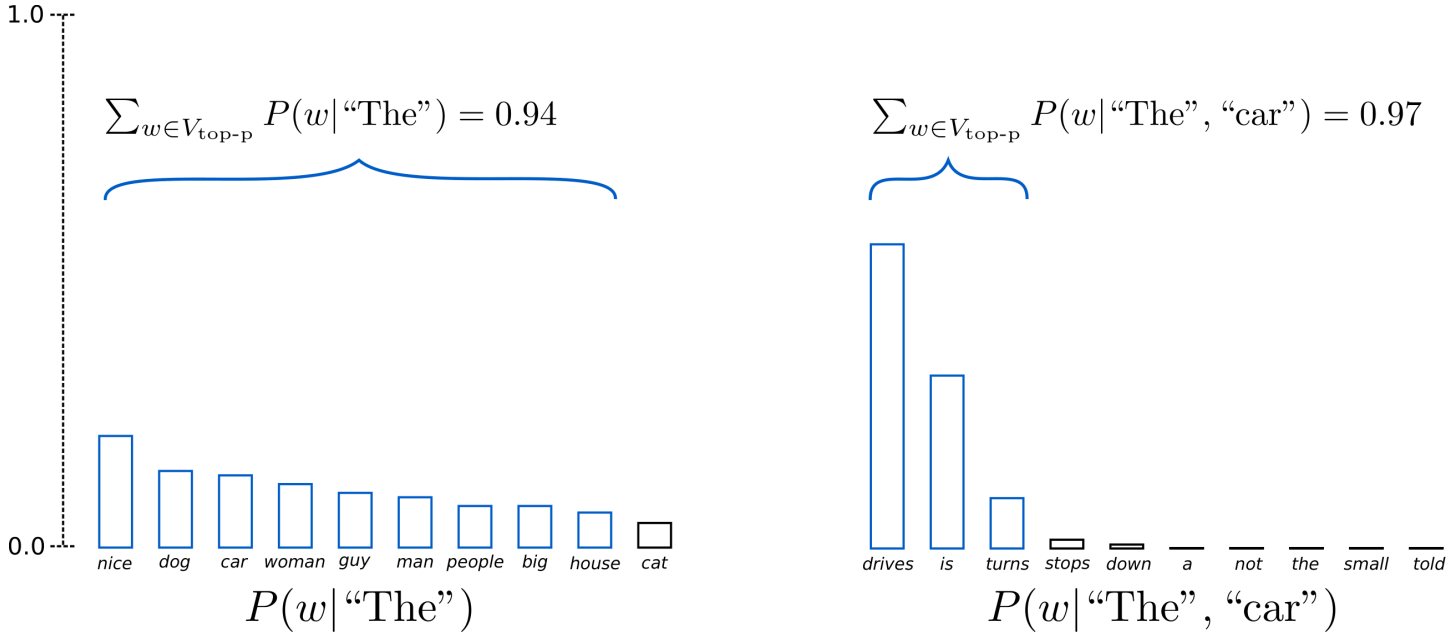
Decoding methods

- Greedy decoding
- Beam search (tree)
- **Top-k sampling**



Decoding methods

- Greedy decoding
- Beam search (tree)
- Top-k sampling
- **Top-p sampling**



Jupyter

- lecture_2_inference.ipynb

Course Projects = Shared Task

1. Create **dataset** [18/03-25/03] (20 samples per student)
   - Prompt: situation / question
   - Correct answer
   - Sources

2. Apply **models** to dataset [25/03-8/04]
   - Prompt LLMs (Inference)
   - Fine-tune models (Training)

3. **Evaluate** model performance [8/04-15/04]
   - Annotate results
   - Calculate metric, e.g., Accuracy

4. **Presentation** [15/04-29/04]
   - Error analysis
   - Report + slides (3)

Dataset: Austrian Tax Law

- Laws
  - EStG 1988 at RIS
  - UStG 1994
  - KStG
  - GrEStG

- Guidelines
  - EStR 2000 at Findok

- ->
  - Choose team
  - Choose topic

Dataset: Austrian Tax Law


```
[
  {
    "id": "VAT-INTL-001",
    "prompt": "Eine schweizerische Gesellschaft verkauft an ein anderes
schweizerisches Unternehmen ein Reisepaket für Unternehmer und verrechnet dabei
auch Eintrittsberechtigungen für kulturelle Veranstaltungen in Österreich weiter.
Die Weiterverrechnung erfolgt im eigenen Namen im Rahmen einer Besorgungsleistung.
Liegt der Leistungsort der weiterverrechneten Eintrittsberechtigungen in
Österreich?",
    “correct_answer": "Ja. Besorgt ein Unternehmer eine sonstige Leistung, sind
nach § 3a Abs. 4 UStG 1994 die für die besorgte Leistung geltenden Vorschriften
entsprechend anzuwenden. Bei Eintrittsberechtigungen zu kulturellen
Veranstaltungen bestimmt sich der Leistungsort nach § 3a Abs. 11a UStG 1994 nach
dem Ort der tatsächlichen Veranstaltung; findet diese in Österreich statt, liegt
der Leistungsort in Österreich.”,
    "sources": [
      "§ 3a Abs. 4 UStG 1994",
      "§ 3a Abs. 11a UStG 1994"
    ]
  }
]
```

1. Annotating Laws (EStG, UStG)

Standard Format: `§ [Number] Abs [Paragraph] Z [Item] lit [Letter] [Law Abbreviation]`
Example: `§ 26 Abs 1 Z 2 lit a EStG 1988`

Breakdown:

- `§ 26`: The main section.
- `Abs 1`: *Absatz* (Paragraph).
- `Z 2`: *Ziffer* (Numerical item in a list).
- `lit a`: *Littera* (Lettered item in a list).
- `EStG 1988`: The short name of the law.

2. Annotating Guidelines (EStR 2000)

Standard Format: `[Guideline Abbreviation] Rz [Number]`

Example: `EStR 2000 Rz 650`

GitHub